

HR ANALYTICS DASHBOARD – EMPLOYEE ATTRITION ANALYSIS

Project Overview: -

This project analyzes employee attrition patterns using the IBM HR Analytics dataset. The objective is to identify the key factors contributing to employee turnover and provide actionable recommendations to improve employee retention.

The dashboard was developed in Microsoft Excel using Pivot Tables, Pivot Charts, KPI Cards, Slicers, and data analysis techniques.

Business Problem: -

Employee attrition can negatively impact productivity, recruitment costs, and organizational performance.

The HR department wants to understand:

- What is the overall attrition rate?
- Which departments experience the highest attrition?
- Which job roles have the highest turnover?
- Does overtime contribute to employee attrition?
- Which age groups are most likely to leave?
- Does employee income influence attrition?

Dataset Information: -

Dataset Name: IBM HR Analytics Employee Attrition Dataset

Source: Kaggle

Records: 1,470 Employees

Columns: 35

Key Fields Used

- Age
- Attrition
- Department
- Gender
- Job Role
- Monthly Income
- Over Time
- Marital Status
- Years At Company
- Performance Rating

Data Cleaning & Preparation: -

The following data quality checks were performed:

Missing Values

- No missing values found.

Duplicate Records

- No duplicate EmployeeNumber records found.

Constant Columns Identified

- EmployeeCount
- Over18
- StandardHours

These columns were excluded from analytical decision-making because they contained constant values.

Additional Columns Created: -

AgeGroup

- Under 30
- 30–40
- 41–50
- 50+

Income Group

- Low Income
- Medium Income
- High Income

Key Performance Indicators (KPIs)

KPI	Value
Total Employees	1470
Employees Left	237
Active Employees	1233
Attrition Rate	16.12%
Average Age	37 Years
Average Monthly Income	6502.93

Analysis Performed: -

Department-wise Attrition

Analyzed employee attrition across departments to identify areas with the highest employee turnover.

Job Role Attrition

Identified job roles experiencing the highest number of employee exits.

Age Group Analysis

Examined attrition trends across different employee age groups.

Overtime Analysis

Evaluated the relationship between overtime work and employee attrition.

Gender Analysis

Compared attrition patterns between male and female employees.

Marital Status Analysis

Studied employee attrition across marital status categories.

Income Analysis

Analyzed attrition patterns across different income groups.

Key Insights: -

1. Research & Development accounts for approximately 56% of total employee attrition.
2. Laboratory Technicians experienced the highest number of employee exits among all job roles.
3. Most employee attrition occurs among employees below the age of 40.
4. Employees working overtime show significantly higher attrition compared to employees who do not work overtime.
5. Approximately 69% of attrition comes from employees in the low-income category.

6. Single employees account for the highest proportion of employee attrition compared to married and divorced employees.

Recommendations: -

1. Conduct retention and engagement programs within the Research & Development department.
2. Review workload distribution and overtime policies to improve work-life balance.
3. Implement career development and mentorship programs for employees under 40 years of age.
4. Review compensation structures for low-income employees to improve retention.
5. Conduct employee satisfaction surveys among Laboratory Technicians and Sales Executives to identify role-specific concerns.

Tools & Techniques Used: -

Microsoft Excel

- Pivot Tables
- Pivot Charts
- KPI Dashboard
- Conditional Formatting
- Slicers
- Data Cleaning
- Business Analysis
- Dashboard Design

Dashboard Features

- Interactive KPI Cards
- Department-wise Attrition Analysis
- Job Role Attrition Analysis
- Age Group Analysis
- Income Group Analysis
- Gender Analysis
- Marital Status Analysis
- Dynamic Filtering using Slicers

Conclusion: -

The analysis reveals that employee attrition is strongly influenced by department, job role, income level, age group, and overtime. Organizations can improve employee retention by focusing on workload management, compensation review, career growth opportunities, and targeted retention strategies for high-risk employee groups.

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